

Introduction To Statistical Mechanics

3. Binomial Distribution

The probability function that we are going to discuss is called the binomial distribution. The reason is that represents a typical term encountered in expanding $(p+q)^N$ by the binomial theorem. Indeed, we recall that the binomial expansion is given by the formula

$$(p+q)^N = \sum_{n=0}^N \frac{N!}{n!(N-n)!} p^n q^{N-n}$$

We already pointed out in earlier that if it is known that the particle has performed n_1 steps to the right in a total of N steps, then its net displacement m from the origin is determined. Thus the probability $P_N(n_1)$ given by the probability function i.e.,

$$P_N(m) = W_N(n_1)$$

$$n_1 = \frac{1}{2}(N+m), \quad n_2 = \frac{1}{2}(N-m)$$

Substitution of these relations

$$P_N(m) = \frac{N!}{\left[\frac{(N+m)}{2}\right]! \left[\frac{(N-m)}{2}\right]!} p^{(N+m)/2} (1-p)^{(N-m)/2}$$

In the special case where $p = q = \frac{1}{2}$ this assumes the symmetrical form

$$P_N(m) = \frac{N!}{\left[\frac{(N+m)}{2}\right]! \left[\frac{(N-m)}{2}\right]!} \left(\frac{1}{2}\right)^N$$

Example: Discuss $N = 3$ and $p = q = \frac{1}{2}$ case.

Solution: Suppose that $p = q = \frac{1}{2}$ and that $N = 3$ as illustrated in **Fig 1.3**. Then the possible numbers of steps to the right are $n_1 = 0, 1, 2, \text{ or } 3$; the corresponding displacements are $m = -3, -1, 1, \text{ or } 3$; the corresponding probabilities are (as is clear from figure)

$$W_3(n_1) = P_3(m) = \frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$$

If we consider the binomial distribution for the same case where $p = q = \frac{1}{2}$, but with the total number of steps $N = 20$. The envelope of these discrete values of $P_N(m)$ is a bell-shaped curve. The physical significance of this is obvious. After N random steps, the probability of the particle being a distance of N steps away from the origin is very small, while the probability of its being located in the vicinity of the origin is largest.

Example: The probability $W(n)$ that an event characterized by a probability p occurs n times in N trials was shown to be given by the binomial distribution

$$W(n) = \frac{N!}{n!(N-n)!} p^n (1-p)^{N-n} \quad (1)$$

Consider a situation where the probability p is small ($p < 1$) and where one is interested in the case $n < N$. (Note that if N is large, $W(n)$ becomes very small if $n \rightarrow N$ because of the smallness of the factor p^n when $p < 1$. Hence $W(n)$ is indeed only appreciable when $n < N$).

Several approximations can then be made to reduce (1) to simpler form.

(a) Using the result $\ln(1-p) \approx -p$, show that $(1-p)^{N-n} \approx e^{-Np}$.

(b) Show that $\frac{N!}{(N-n)!} \approx N^n$

(c) Hence show that (1) reduces to

$$W(n) = \frac{\lambda^n}{n!} e^{-\lambda} \quad (2)$$

where $\lambda = Np$ is the mean number of events. The distribution (2) is called the “Poisson distribution.”

Solution: (a) $\ln(1-p)^{N-n} \approx -p(N-n) \approx -Np$ $n \ll \bar{N}$ thus $(1-p)^{N-n} \approx e^{-Np}$

(b) $\frac{N!}{(N-n)!} = \frac{N(N-1)\cdots(N-n+1)(N-n)!}{(N-n)!} = N(N-1)\cdots(N-n+1) \approx N^n$ if $n \ll N$

(c) $W(n) = \frac{N!}{n!(N-n)!} p^n (1-p)^{N-n} \approx \frac{N^n}{n!} p^n e^{-Np} \approx \frac{\lambda^n}{n!} e^{-\lambda}$, where $\lambda = Np$